Dr. Shanu Dengre

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Summary

AI / ML Engineer with 4.5+ years in Machine Learning Operations (MLOps), shipping production-ready deep learning models to 300+ locations. Builds scalable AI pipelines on Docker/Kubernetes with MLflow lifecycle management, plus generative AI (GenAI) agents and RAG chatbots. PhD in Physics.

Skills

MLOps & Monitoring: MLflow Model Registry, Drift Detection, Grafana, Sentry
Containerization & Orchestration: Docker, Kubernetes, Airflow
Software Engineering: Python, FastAPI, Pydantic, Git/GitLab, CI/CD (GitHub Actions)
Cloud Platforms & AI Services: AWS (SageMaker, S3, Lambda), Azure, GCP, LLM APIs
Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, LSTM/GRU, Hugging Face Transformers
GenAI / LLMs: LangChain, LangGraph, MCP, Agentic Workflows, Tool Calling, RAG, Chatbots
Computer Vision & Image Processing: OpenCV, YOLO-family, RTMDet, SAM, CLIP (Embeddings)
Data Engineering: SQL, PySpark, Pandas, NumPy, MongoDB, Snowflake, Kafka
Documentation & Reporting: Confluence, PowerBI, Plotly, Streamlit, Gradio
Time Series Forecasting: DARTS, TimesFM, CHRONOS, Fine-tuning Transformer Models
AI-Assisted Development: Claude, Gemini, GitHub Copilot
Languages: English (Fluent), German (A2.2), Hindi (Native)

Experience

InterProductTec – Mumbai *Nov 2025 – Present*
Sr. Data Scientist
Demand Forecast Project Lead

- Owned the forecast project end to end, from planning to production, with cross-functional teams.
- Deployed production-ready POS integration pipelines that automate customer onboarding.
- Achieved negligible downtime and SOTA forecast accuracy across 300+ locations in 10+ chains by strengthening production monitoring and reliability (Sentry, PowerBI dashboards).
- Managed forecasting model deployment and lifecycle management with MLflow across 300+ locations.
- Built an MCP-based monitoring tool that raises alerts when forecast accuracy drifts.
- Applied GenAI via LangChain/LangGraph agents that auto-link restaurant recipes to POS items.
- Improved holiday/offer sales forecasts by 20% with XGBoost/LightGBM and deep learning.
- Cut food waste by 15% and improved crew efficiency by 20% through demand forecasting.
- Automated food waste reports as reusable, documented Python modules, cutting manual effort by 50%.
- Mentored junior data scientists via code reviews and pairing; documented pipelines and best practices.

PreciTaste – Munich *Feb 2022 – Oct 2025*
AI Engineer and Data Scientist
Time Series Forecasting – Sales and Demand

- Automated DARTS forecasting and retraining with drift detection, cutting retraining time by 50%.
- Cut query latency by 60% with containerized FastAPI microservices (Docker, Kubernetes, MongoDB).

Computer Vision – Object Detection and Segmentation

- Trained (MLflow) and deployed YOLO/PyTorch neural networks for food recognition in 5+ chains.
- Built automated image-processing annotation pipelines (OpenCV, SAM), cutting labeling time by 50%.
- Built real-time POS order matching on live Kafka: 95%+ validation accuracy, <100ms latency.

Live Demos

AI News Navigator | hf.co/spaces/shanudengre82/ai-news-navigator *RAG Chatbot, LangGraph*
Next Restaurant Recommender | next-restaurant.onrender.com *NLP, Embeddings*
Newsvendor Problem Visualiser | newsvendorproblem.streamlit.app *Python, Streamlit*

Education

PhD (Physics), TU Dresden | Publication: Phys. Rev. B 103, 064425 (2021) Oct 2015 – Feb 2021
BSc–MSc (Physics), IISER Pune | Publication: Phys. Rev. B 95, 054401 (2017) Aug 2010 – May 2015